

JUNGWOO PARK

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EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY, Atlanta, GA Aug. 2024–Present
Master's in Computer Science, College of Computing

UNIVERSITY OF CALIFORNIA SAN DIEGO (UCSD), San Diego, CA Sep. 2019–June 2024
Bachelor of Science in Cognitive Science (Specialization in Machine Learning and Neural Computation)

RESEARCH EXPERIENCE

ML Models Team Research Assistant, Department of Computer Science, Georgia Tech, GA Jan. 2025–Present

PopSign: Sign Language Learning Games and Technologies

- Preprocessed and analyzed the phrase data for training/testing and eventual implementation in the CopyCat game
- Currently building ML pipeline to do the Seq2Seq phrase to sign matching task and implement model conversion
- <https://www.popsign.org/>

PROJECTS

Personalized AI Author Sep. 2025–Nov. 2025

- Developed a user tailored GPT-5 nano single repetitive agent that plans, drafts, edits, critiques, and rewrites personalized books/essays using ReAct prompting and Project Gutenberg corpus for RAG
- https://github.com/pjw0417/Personalized_AI_Author

Reimplementation of Denoising Diffusion Probabilistic Models in PyTorch Aug. 2025–Sep. 2025

- Reimplemented DDPM paper from scratch in PyTorch and trained diffusion model on CIFAR-10 dataset
- Achieved FID score of 39.28 with 10,000 target steps and 10,000 samples
- https://github.com/pjw0417/ddpm_pytorch

A RAG-Powered LLM for Personalized Restaurant Recommendation in Atlanta May 2025–Aug. 2025

- Developed a personalized restaurant recommender for Atlanta using RAG, LLaMA 3, and prompt engineering, combining structured metadata with natural language reviews.
- <https://github.com/pjw0417/RAGaurant>

Exploring effectiveness of deep neural nets for solving Korean word spacing problem Jan. 2025–Apr. 2025

- Designed and compared the performances of Bidirectional LSTM, CNN, and Transformer based natural language processing model using PyTorch on Korean word spacing problem
- <https://github.com/kyuyeonlee/nlp-kiss/blob/main/Final%20Report.pdf>

Detecting User Actions from Mouse Events Aug. 2024–Dec. 2024

- Engineered ML pipeline using mouse movement logs to classify real-time user activity of varying window sizes
- Compared performances of LightGBM, Random Forest, KNN, LSTM, and K-Means
- <https://github.com/pjw0417/ML-Fall24-Team7/blob/main/docs/FINAL.md>

AWARDS

Republic of Korea (ROK) Army, Chuncheon, Korea Jan. 2021–Jul. 2022

- Won 1st Place and Grand Prize in multiple ROK Army start-up competitions; ranked Top 25 out of 512 teams nationally (Sep-Nov. 2021)

SKILLS

Technical skills: Python, Java, PyTorch, scikit-learn, Machine Learning, Deep Learning, Git, Hugging Face, Fine Tuning, Prompt Engineering, RAG, LLM Agent, LangChain